Miss Julie: Our Own AI Teaching Engine

A practical redesign proposal for Talkora • KG to Class 10 • Presentation edition

FROM API-DEPENDENT CHAT TO A CONTROLLED AI TUTOR

One shared open-source language model, guided by Talkora's curriculum, student level, and teaching rules. Speech and animations remain separate services.

LEARN

Syllabus-mapped lessons, adaptive practice, age-aware prompts

SPEAK

Miss Julie voice, gentle music, student turn-taking

MEASURE

Progress, homework, teacher and principal reports

110 LEVELS

11 GRADES

ONE SHARED MODEL

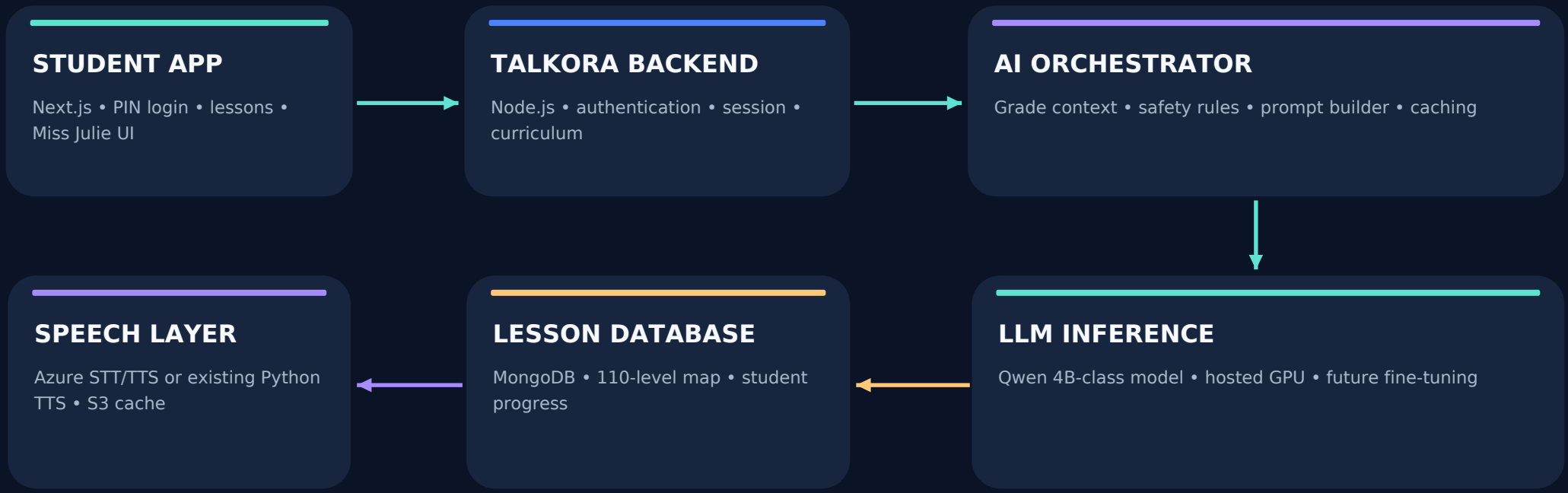
PROVIDER SWITCHING

PRESENTATION GOAL

Demonstrate a friendly bilingual teacher in a free Colab prototype, then move to a secure, scalable school pilot without rebuilding the existing Talkora frontend.

System architecture

How the proposed AI fits the existing Talkora application



A RESPONSE FLOW

1. Student speaks → 2. Speech-to-text → 3. Backend sends lesson context to Qwen → 4. Guardrails check response → 5. Voice plays; music ducks → 6. Progress saved.

Security: LLM endpoint stays server-side. Never expose API keys or private student data to the browser. Keep provider fallback and request limits.

Free prototype: what sir will see

A realistic proof-of-concept before we spend on infrastructure

1 CHOOSE A GRADE

KG: letters and simple words. Class 1: short sentences. Class 4: comprehension and dialogue.

2 TALK TO MISS JULIE

Friendly, bilingual English/Hindi responses with gentle corrections and follow-up questions.

3 CHECK LEARNING

Show sample feedback, adaptive hints, and a short spoken-homework prompt.

LIVE DEMO SCRIPT

Student (KG): "Miss, what is A?"

Miss Julie: "A is for apple! Can you say apple? Great! Now find something that starts with A."

Student (Class 4): "Can you help me describe my school?"

Miss Julie: "Of course! Start with: My school is... Tell me one thing you like about it."

PROTOTYPE STACK

Google Colab (free GPU when available) + quantized Qwen + teaching prompts + simple Gradio chat UI. Text-first; not yet a production service.

Implementation roadmap

Upgrade the current project in controlled, testable stages

PHASE 01

Prototype

Colab notebook, KG/Class 1/Class 4 personas, English/Hindi samples, evaluation rubric.

PHASE 02

Backend adapter

Provider interface, Qwen endpoint client, configurable .env, timeouts, fallback.

PHASE 03

Teaching intelligence

Grade-specific curriculum retrieval, adaptive hints, safe responses, homework.

PHASE 04

Voice & UX

Keep existing voice/S3 flow; background music separate; pause for microphone.

PHASE 05

School pilot

GPU hosting, load tests, logging, student privacy, cost controls, rollout.

Production cost & decision

Planning ranges, not live vendor quotations • benchmark before commitment

COST MODEL

Monthly AI compute ≈ billable GPU hours × GPU hourly rate. For scale-to-zero services, also allow for cold starts, minimum billing, storage and request overhead.

Example only: 80 billable GPU hours / month

Budget rented GPU Lowest raw price; idle rental can still bill.	**$0.25–$0.50/hr**	**$20–$40**
On-demand GPU Better for intermittent school sessions.	**$0.60–$1.20/hr**	**$48–$96**
Managed inference API Simpler launch; benchmark total cost.	**Token-based**	**Usage-dependent**

NOT INCLUDED: speech recognition, voice generation, S3, backend, database, egress, taxes, or staff time.

GO / NO-GO: verify teacher quality, Hindi-English accuracy, response latency, concurrency, and per-student cost.